



Chapter 2

Data Models



Learning Objectives

- After completing this chapter, you will be able to:
 - Discuss data modeling and why data models are important
 - Describe the basic data-modeling building blocks
 - Define what business rules are and how they influence database design
 - Understand how the major data models evolved
 - List emerging alternative data models and the needs they fulfill
 - Explain how data models can be classified by their level of abstraction



Data Modeling and Data Models

- Data modeling: creating a specific data model for a determined problem domain
 - Data model: simple representation of complex real-world data structures
 - Useful for supporting a specific problem domain
 - Model: abstraction of a more complex real-world object or event



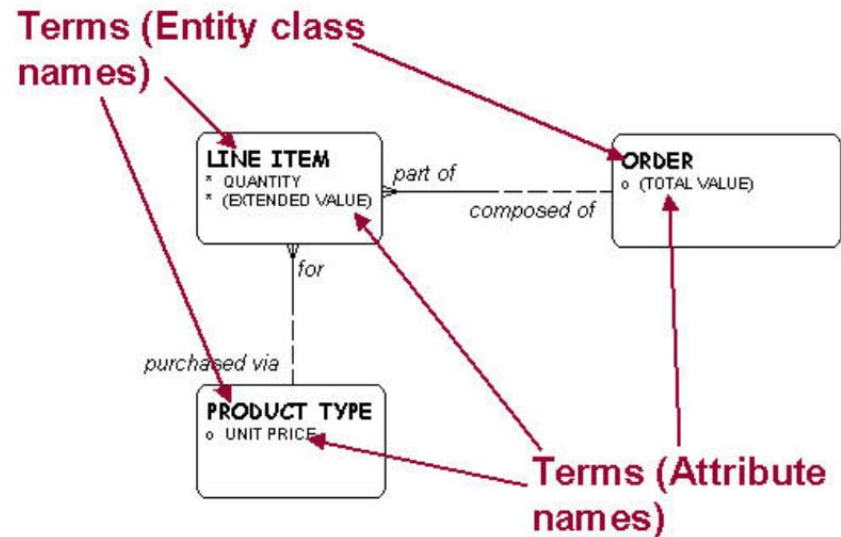
The Importance of Data Models

- The importance of data modeling cannot be overstated
 - Facilitates communication
 - Gives various views of the database
 - Organizes data for various users
 - Provides an abstraction for the creation of good a database



Data Model Basic Building Blocks

- Entity: person, place, thing, or event about which data will be collected and stored
 - Attribute: characteristic of an entity
 - Relationship: association among entities
 - One-to-many (1:M OR 1..*)
 - Many-to-many (M:N or *..*)
 - One-to-one (1:1 OR 1..1)
- Constraint: restriction placed on data
 - Ensures data integrity



<https://tdan.com/modeling-business-rules-what-data-models-do/5174>

1. Business rule: No *single* customer order may be less than \$1.00 or exceed \$500.00.
2. Business rule: No high-school instructor can sponsor more than two student committees.
3. Constraint: Only MasterCard, VISA, and Discover button options appear on an order form.

<https://www.relationaldbdesign.com/database-design/module4/business-rules-exercise.php>



Business Rules

- Brief, precise, and unambiguous description of a policy, procedure, or principle
 - Create and enforce actions within that organization's environment
 - Establish entities, relationships, and constraints

The following statement is an example of a typical business rule:

A SHIP DATE cannot be prior to an ORDER DATE for any given order.

Here is an example of a typical application oriented business rule:

A customer with a "Preferred" status receives a 15% discount on all purchases.

An instructor must teach one class, but no more than eight classes.



Discovering Business Rules (1 of 2)

- Sources of business rules
 - Company managers
 - Policy makers
 - Department managers
 - Written documentation
 - Direct interviews with end users



Discovering Business Rules (2 of 2)

- Reasons for identifying and documenting business rules
 - Standardize company's view of data
 - Facilitate communications tool between users and designers
 - Assist designers
 - Understand the nature, role, scope of data, and business processes
 - Develop appropriate relationship participation rules and constraints
 - Create an accurate data model



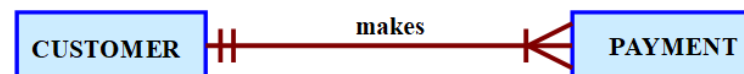
Translating Business Rules into Data Model Components

- Business rules set the stage for the proper identification of entities, attributes, relationships, and constraints
 - Nouns translate into entities
 - Verbs translate into relationships among entities
- Relationships are bidirectional
 - Questions to identify the relationship type
 - How many instances of B are related to one instance of A?
 - How many instances of A are related to one instance of B?

Chen model



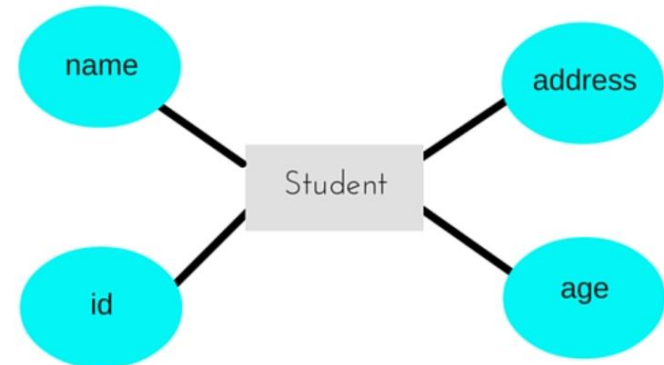
Crow's Foot model





Naming Conventions

- Entity name requirements
 - Be descriptive of the objects in the business environment
 - Use terminology that is familiar to the users
- Attribute name
 - Required to be descriptive of the data represented by the attribute
- Proper naming
 - Facilitates communication between parties
 - Promotes self-documentation



<https://www.studytonight.com/dbms/database-model.php>



Break out room: Business Rules Example (1 of 2)

For each stated business rule, draw the data model including entity name and relationship type using crow's feet notation.

- An invoice contains one or more invoice lines.
- Each invoice line is associated with a single invoice.
- A store employs many employees.
- Each employee is employed by only one store
- A college has many departments.
- Each department belongs to a single college.

<https://www.chegg.com/homework-help/questions-and-answers/database-fundamentals-class-required-15-20-business-rules-task-2--also-need-er-diagram-tas-q31442952>



Break out room: Business Rules Example (2 of 2)

For each stated business rule, draw the data model including entity name and relationship type using crow's feet notation.

- A driver may be assigned to drive many different vehicles.
- Each vehicle can be driven by many drivers. (Note: Keep in mind that this business rule reflects the assignment of drivers during some period of time.)
- A client may sign many contracts.
- Each contract is signed by only one client.
- A sales representative may write many contracts.
- Each contract is written by one sales representative.

<https://www.chegg.com/homework-help/questions-and-answers/database-fundamentals-class-required-15-20-business-rules-task-2--also-need-er-diagram-tas-q31442952>



Business Rules Example with Constraints

- Every employee should be present by 9am and work 9 hours per day.
- Every employee must under do a MATH course in order to get an increase salary.
- The company will offer a bonus of half a salary in April.
- Every student must have 80% attendance in order to allow for taking exam.
- A student has to obtain 60 out of 100 in Workshop subject for passing.

Adjusted from : <https://www.slideshare.net/TharinduWeerasinghe/business-rules-in-databases-125026916>



Hierarchical and Network Models (1 of 2)

- Hierarchical models: developed to manage large amounts of data for complex manufacturing projects
 - Represented by an upside-down tree which contains segments
 - Segments are the equivalent of a file system's record type
 - Depicts a set of one-to-many (1:M) relationships



Hierarchical and Network Models (2 of 2)

- Network models: created to represent complex data relationships effectively
 - Improved database performance and imposed a database standard
 - Allows a record to have more than one parent
- Standard database concepts that emerged with the network model are still used by modern data models
 - Schema and subschema
 - Data manipulation language (DML)
 - Data definition language (DDL)



The Relational Model (1 of 4)

- Produced an automatic transmission database that replaced standard transmission databases
 - Based on a relation (i.e., table): matrix composed of intersecting tuples (rows) and attributes (columns)
- Describes a precise set of data manipulation constructs



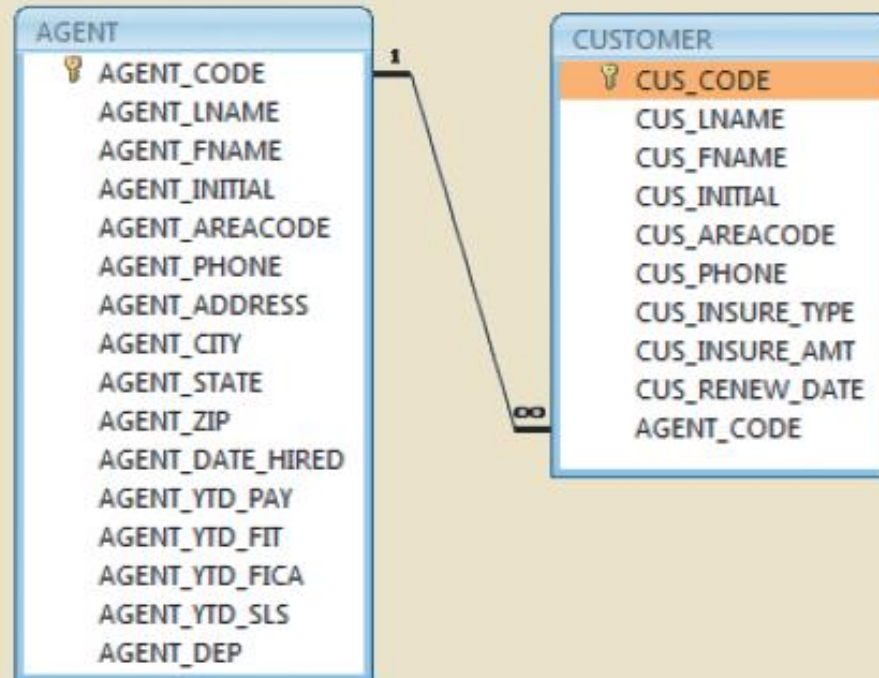
The Relational Model (2 of 4)

- Relational database management system (RDBMS)
 - Performs basic functions provided by the hierarchical and network DBMS systems
 - Makes the relational data model easier to understand and implement
 - Hides the complexities of the relational model from the user



The Relational Model (3 of 4)

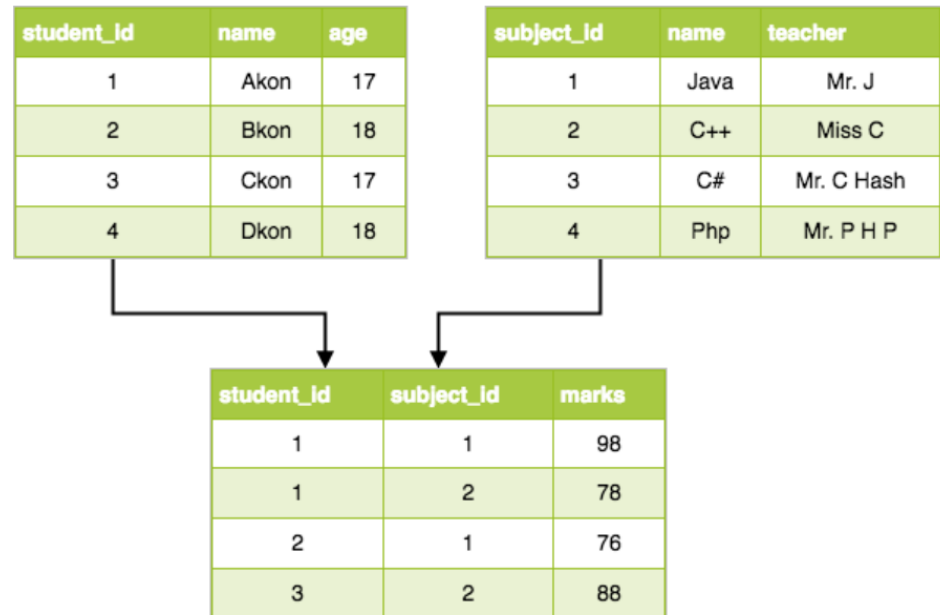
FIGURE 2.2 A RELATIONAL DIAGRAM





The Relational Model (4 of 4)

- SQL-based relational database application
 - End-user interface
 - Allows end user to interact with the data
 - Collection of tables stored in the database
 - Each table is independent from another
 - Rows in different tables are related based on common values in common attributes
 - SQL engine
 - Executes all queries





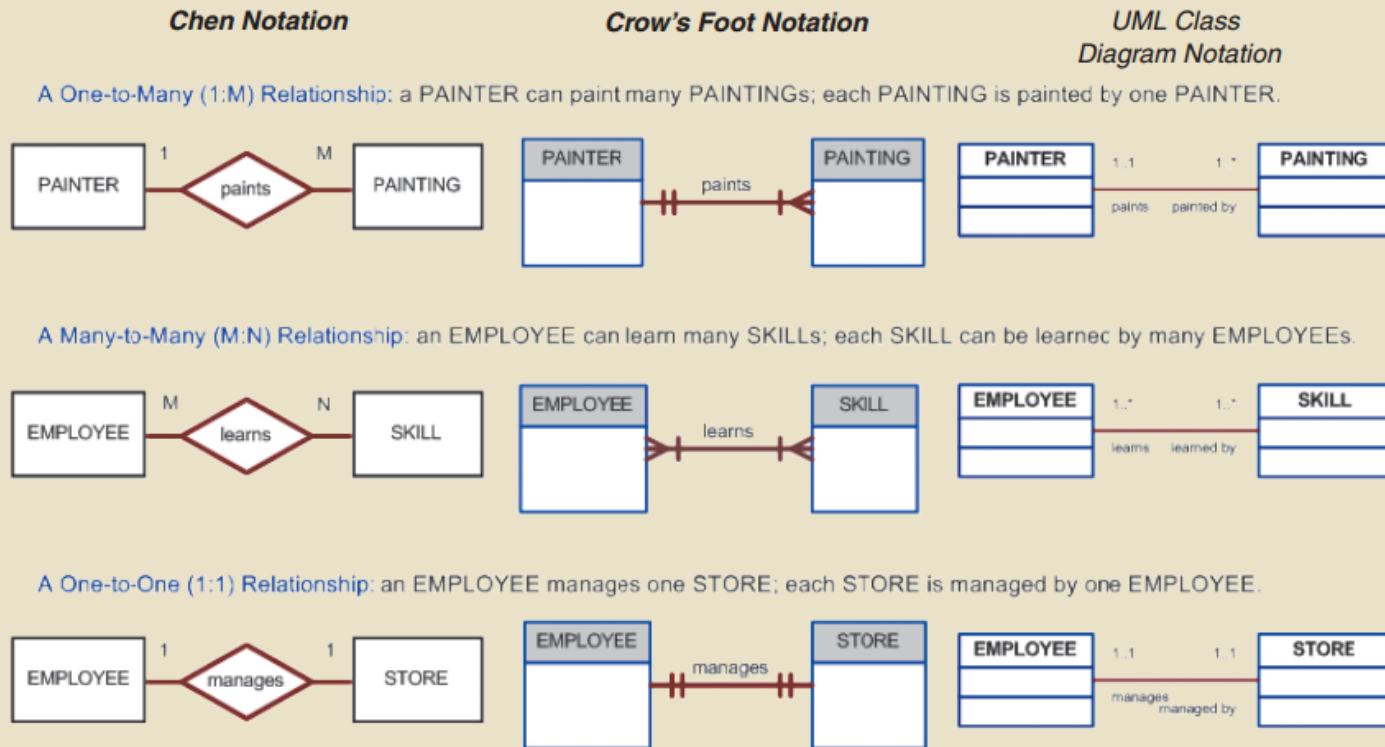
The Entity Relationship Model (1 of 2)

- Graphical representation of entities and their relationships in a database structure
 - Entity relationship diagram (ERD): uses graphic representations to model database components
 - Entity instance or entity occurrence: rows in the relational table
 - Attributes: describe particular characteristics
 - Connectivity: term used to label the relationship types



The Entity Relationship Model (2 of 2)

FIGURE 2.3 THE ER MODEL NOTATIONS





The Object-Oriented Data Model (1 of 3)

- Both data and its relationships are contained in a single structure known as an object
 - Object-oriented database management system(OODBMS): based on OODM
- Object: contains data and their relationships with operations that are performed on it
 - Basic building block for autonomous structures
 - Abstraction of real-world entity
- Attribute: describes the properties of an object



The Object-Oriented Data Model (2 of 3)

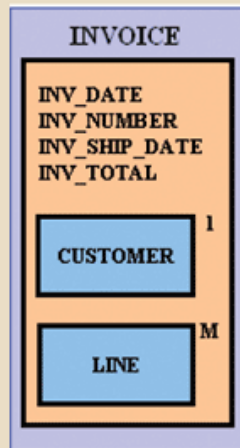
- Class: collection of similar objects with shared structure and behavior organized in a class hierarchy
- Class hierarchy: resembles an upside-down tree in which each class has only one parent
- Inheritance: object inherits methods and attributes of classes above it
- Unified Modeling Language (UML): describes sets of diagrams and symbols to graphically model a system



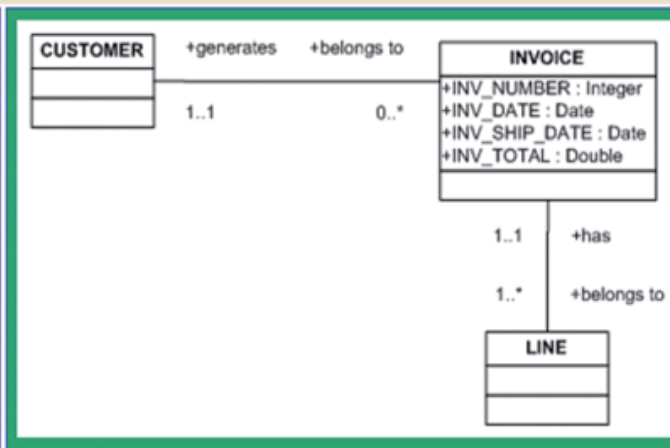
The Object-Oriented Data Model (3 of 3)

FIGURE 2.4 A COMPARISON OF THE OO, UML, AND ER MODELS

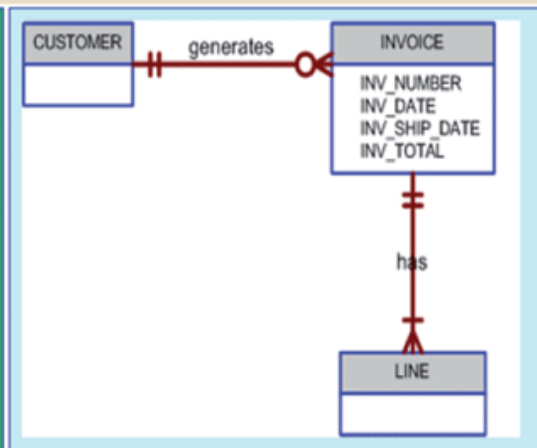
Object Representation



UML Class Diagram



ER Model





Object/Relational and XML

- Extended relational data model (ERDM)
 - Supports OO features, extensible data types based on classes, and inheritance
 - Object/relational database management system (O/R DBMS): based on ERDM
- Extensible Markup Language (XML)
 - Manages unstructured data for efficient and effective exchange of structured, semistructured, and unstructured data



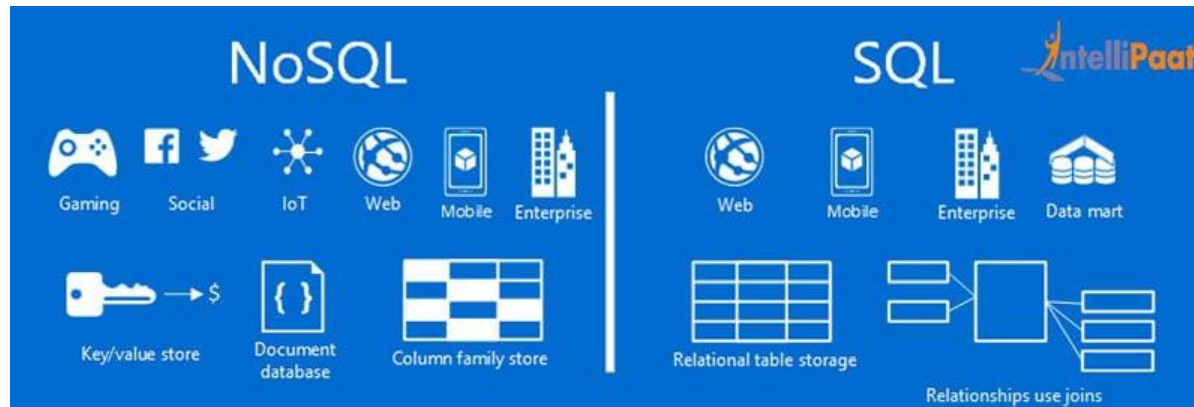
Emerging Data Models: Big Data and NoSQL (1 of 3)

- Goals of Big Data
 - Find new and better ways to manage large amounts of web and sensor-generated data
 - Provide high performance at a reasonable cost
- Characteristics of Big Data
 - Volume
 - Velocity
 - Variety



Emerging Data Models: Big Data and NoSQL (2 of 3)

- Challenges of Big Data
 - Volume doesn't allow usage of conventional structures
 - Expensive
 - OLAP tools proved inconsistent dealing with unstructured data
- New technologies of Big Data
 - Hadoop
 - Hadoop Distributed File System (HDFS)
 - MapReduce
 - NoSQL

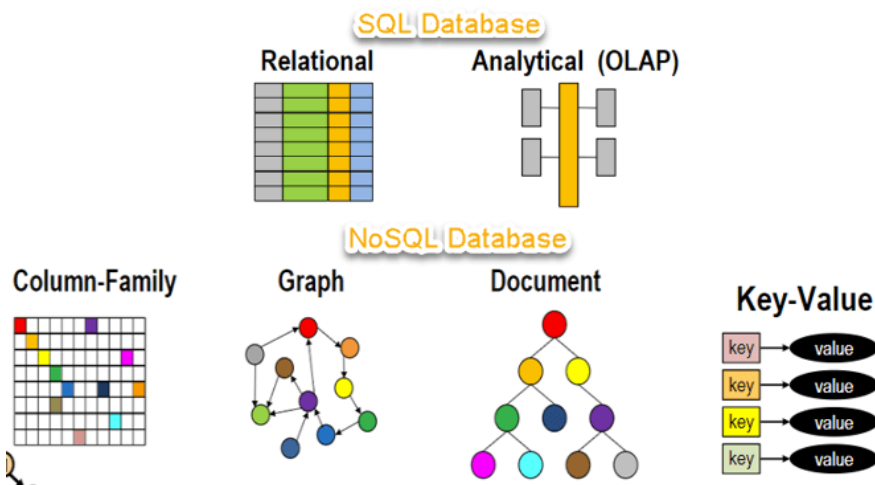


https://medium.com/@sangeethraaj_70563/sql-vs-nosql-faef10e3852d



Emerging Data Models: Big Data and NoSQL (3 of 3)

- NoSQL databases
 - Not based on the relational model
 - Support distributed database architectures
 - Provide high scalability, high availability, and fault tolerance
 - Support large amounts of sparse data
 - Geared toward performance rather than transaction consistency
 - Provides a broad umbrella for data storage and manipulation



<https://www.guru99.com/nosql-tutorial.html>



SQL VS NoSQL Queries

NoSQL Query:

```
db.users.find(  
  { age: { $gt: 18 } },  
  { name: 1, address: 1 }  
) .limit(5)
```

← collection
← query criteria
← projection
← cursor modifier

SQL Query:

```
SELECT _id, name, address  
FROM users  
WHERE age > 18  
LIMIT 5
```

← projection
← table
← select criteria
← cursor modifier

https://medium.com/@sangeethraaj_70563/sql-vs-nosql-faef10e3852d



Clinic-R-Us

Data stored using relational model

PAT_NUM	PAT_LNAME	PAT_FNAME	PAT_DOB
100	Kolmycz	George	15-Jun-1942
101	Lewis	Rhonda	19-Mar-2005
102	Vandam	Rhett	14-Nov-1958
103	Jones	Anne	16-Oct-1974
104	Lange	John	08-Nov-1971
105	Williams	Robert	14-Mar-1975
106	Smith	Jeanine	12-Feb-2003
107	Diante	Jorge	21-Aug-1974
108	Wiesenbach	Paul	14-Feb-1966
109	Smith	George	18-Jun-1961
110	Genkazi	Leighla	19-May-1970
111	Washington	Rupert	03-Jan-1966
112	Johnson	Edward	14-May-1961
113	Smythe	Melanie	15-Sep-1970
114	Brandon	Marie	02-Nov-1932
115	Saranda	Hermine	25-Jul-1972
116	Smith	George	08-Nov-1965

Patient Data

Data stored using key-value model

PAT_NUM	KEY	VALUE
100	PAT_LNAME	Kolmycz
100	PAT_FNAME	George
100	PAT_DOB	15-Jun-1942
101	PAT_LNAME	Lewis
101	PAT_FNAME	Rhonda
101	PAT_DOB	19-Mar-2005
102	PAT_LNAME	Vandam
102	PAT_FNAME	Rhett
102	PAT_DOB	14-Nov-1958

Sample data...

TEST_ID	TEST_DATE	PAT_NUM	HDL_CHOL	LDL_CHOL	TRIGLYCER	GLUCOSE
1	5/23/2009	100	45	120	154	
2	11/13/2010	100	44	121		
3	6/14/2009	101	54	129		108
5	9/24/2010	101	50	123		112
6	8/1/2011	101	52	122		106
7	7/23/2009	102	40	115	135	
8	7/15/2010	102	39	118	139	
9	8/17/2009	103		134	159	118
10	6/15/2010	103		128	153	113

Test Data

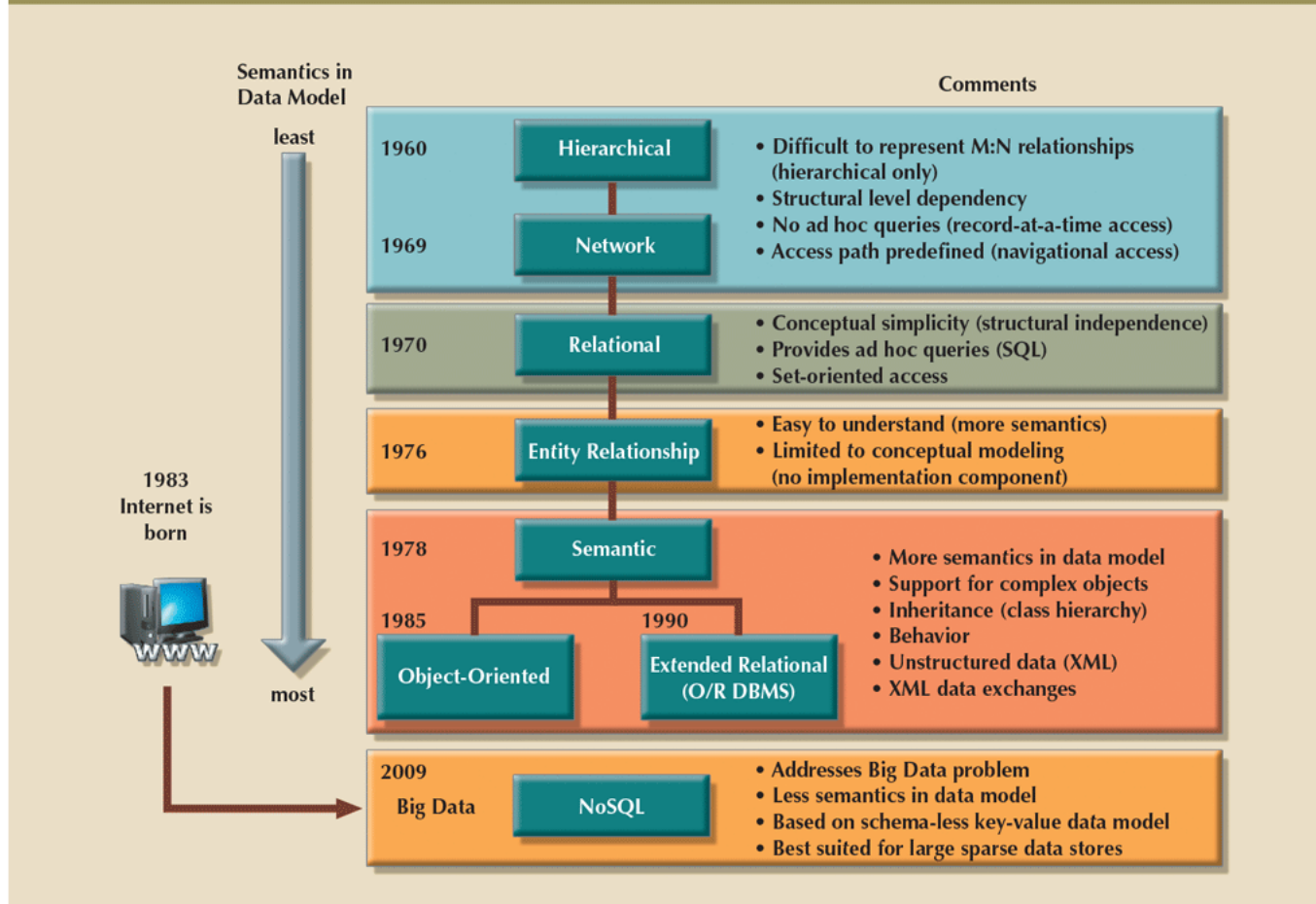
TEST_ID	KEY	VALUE
1	TEST_DATE	5/23/2009
1	PAT_NUM	100
1	HDL_CHOL	45
1	LDL_CHOL	120
1	TRIGLYCER	154
2	TEST_DATE	11/13/2010
2	PAT_NUM	100
2	HDL_CHOL	44
2	LDL_CHOL	121

Sample data...



Data Models: A Summary

FIGURE 2.5 THE EVOLUTION OF DATA MODELS





Hierarchical Model

- Advantages
 - Promotes data sharing
 - Parent/child relationship promotes conceptual simplicity and data integrity
 - Database security is provided and enforced by DBMS
 - Efficient with 1:M relationships
- Disadvantages
 - Requires knowledge of physical data storage characteristics
 - Navigational system requires knowledge of hierarchical path
 - Changes in structure require changes in all application programs
 - Implementation limitations
 - No data definition
 - Lack of standards



Network Model

- Advantages
 - Conceptual simplicity
 - Handles more relationship types
 - Data access is flexible
 - Data owner/member relationship promotes data integrity
 - Conformance to standards
 - Includes data definition language (DDL) and data manipulation language (DML)
- Disadvantages
 - System complexity limits efficiency
 - Navigational system yields complex implementation, application development, and management
 - Structural changes require changes in all application programs



Relational Model

- Advantages
 - Structural independence is promoted using independent tables
 - Tabular view improves conceptual simplicity
 - Ad hoc query capability is based on SQL
 - Isolates the end user from physical-level details
 - Improves implementation and management simplicity
- Disadvantages
 - Requires substantial hardware and system software overhead
 - Conceptual simplicity gives untrained people the tools to use a good system poorly
 - May promote information problems



Entity Relationship Model

- Advantages
 - Visual modeling yields conceptual simplicity
 - Visual representation makes it an effective communication tool
 - Is integrated with the dominant relational model
- Disadvantages
 - Limited constraint representation
 - Limited relationship representation
 - No data manipulation language
 - Loss of information content occurs when attributes are removed from entities to avoid crowded displays



Object-Oriented Model

- Advantages
 - Semantic content is added
 - Visual representation includes semantic content
 - Inheritance promotes data integrity
- Disadvantages
 - Slow development of standards caused vendors to supply their own enhancements
 - Complex navigational system
 - Learning curve is steep
 - High system overhead slows transactions



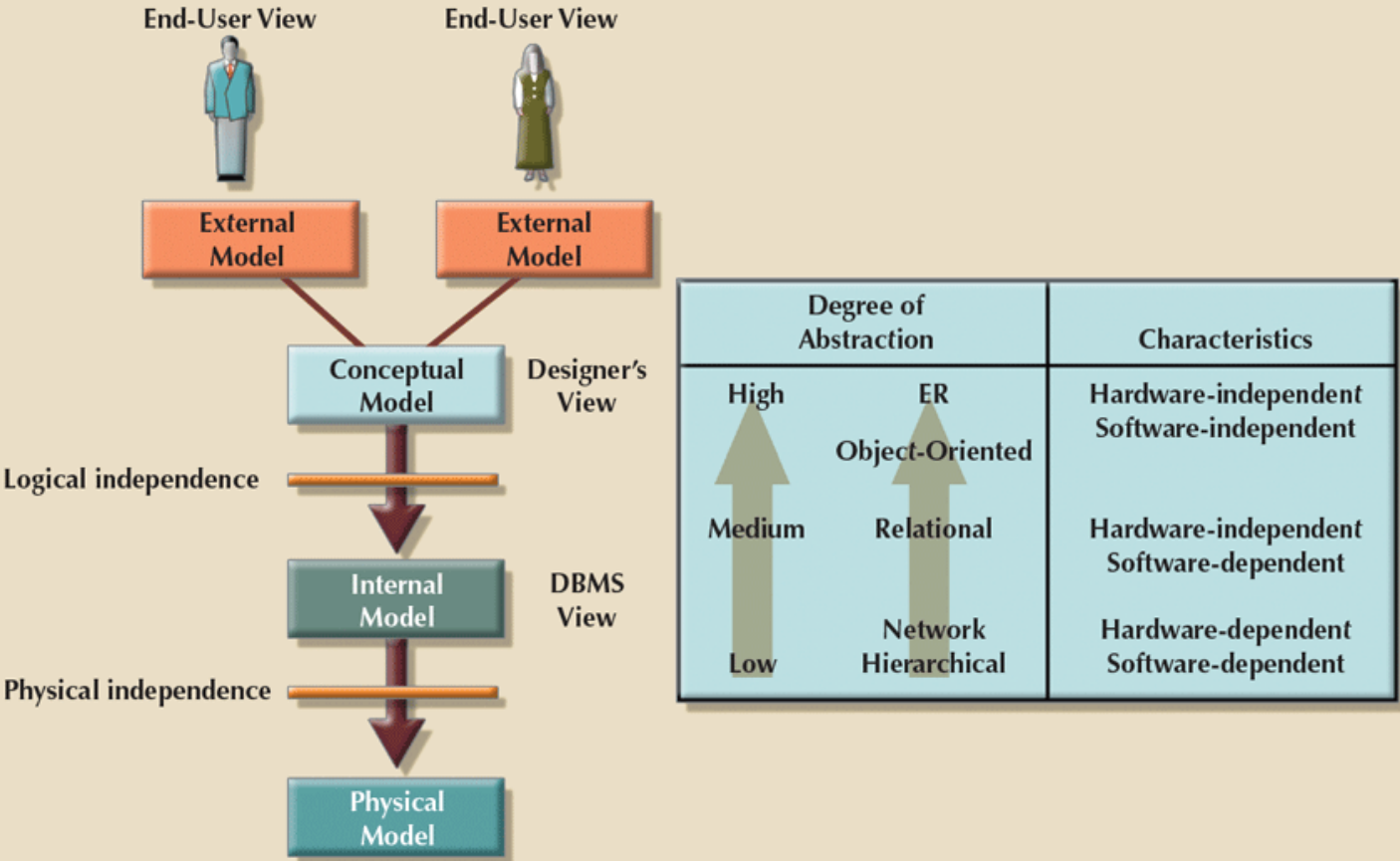
NoSQL

- Advantages
 - High scalability, availability, and fault tolerance are provided
 - Uses low-cost commodity hardware
 - Supports Big Data
 - Key-value model improves storage efficiency
- Disadvantages
 - Complex programming is required
 - There is no relationship support
 - There is no transaction integrity support
 - In terms of data consistency, it provides an eventually consistent model



Degrees of Data Abstraction

FIGURE 2.6 DATA ABSTRACTION LEVELS





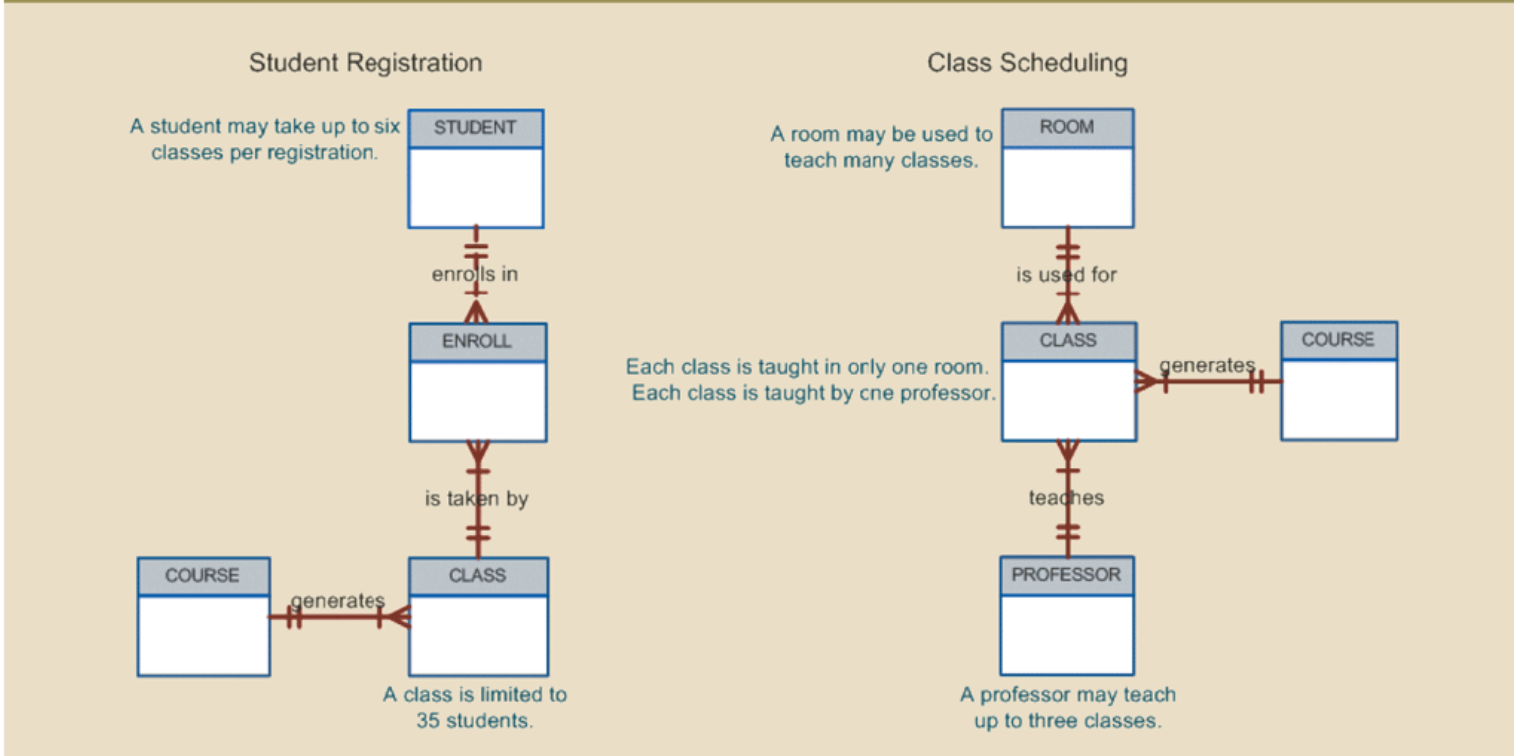
The External Model (1 of 2)

- End users' view of the data environment
 - People who use the application programs to manipulate the data and generate information
- ER diagrams are used to represent the external views
 - External schema: specific representation of an external view



The External Model (2 of 2)

FIGURE 2.7 EXTERNAL MODELS FOR TINY COLLEGE





The Conceptual Model (1 of 2)

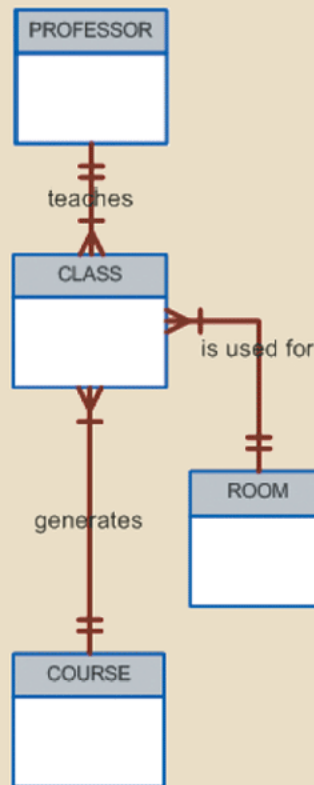
- Represents a global view of the entire database by the entire organization
 - Conceptual schema: basis for the identification and high-level description of the main data objects
 - Logical design: task of creating a conceptual data model
- Conceptual model advantages
 - Macro-level view of data environment
 - Software and hardware independent



The Conceptual Model (2 of 2)

FIGURE 2.9 INTERNAL MODEL FOR TINY COLLEGE

CONCEPTUAL MODEL



INTERNAL MODEL

Create Table PROFESSOR(
PROF_ID NUMBER PRIMARY KEY,
PROF_LNAME CHAR(15),
PROF_INITIAL CHAR(1),
PROF_FNAME CHAR(15),
.....);

Create Table CLASS(
CLASS_ID NUMBER PRIMARY KEY,
CRS_ID CHAR(8) REFERENCES COURSE,
PROF_ID NUMBER REFERENCES PROFESSOR,
ROOM_ID CHAR(8) REFERENCES ROOM,
.....);

Create Table ROOM(
ROOM_ID CHAR(8) PRIMARY KEY,
ROOM_TYPE CHAR(3),
.....);

Create Table COURSE(
CRS_ID CHAR(8) PRIMARY KEY,
CRS_NAME CHAR(25),
CRS_CREDITS NUMBER,
.....);



The Internal Model

- Representing database as seen by the DBMS mapping conceptual model to the DBMS
 - Internal schema: specific representation of an internal model, using the database constructs supported by the chosen database
 - Logical independence: changing internal model without affecting the conceptual model
 - Hardware independent: unaffected by the type of computer on which the software is installed



The Physical Model (1 of 2)

- Operates at lowest level of abstraction
 - Describes the way data are saved on storage media such as magnetic, solid state, or optical media
- Requires the definition of physical storage and data access methods
 - Software and hardware dependent
- Relational model aimed at logical level
 - Does not require physical-level details
- Physical independence: changes in physical model do not affect internal model



The Physical Model (2 of 2)

Table 2.4 Levels of Data Abstraction			
Model	Degree of Abstraction	Focus	Independent of
External	High	End-user views	Hardware and software
Conceptual	Medium-High	Global view of data (database model independent)	Hardware and software
Internal	Medium-Low	Specific database model	Hardware
Physical	Low	Storage and access methods	Neither hardware nor software



Summary

- A data model is an abstraction of a complex real-world data environment
- There are many types of data models (e.g., hierarchical, network, relational, object-oriented, extended relational data model, etc.)
- Data-modeling requirements are a function of different data views (global versus local) and the level of data abstraction